

Capstone Project Details -

Project Instructions for Students

Team Composition

- Each project must be completed in **teams of 2 members**.
 - Teams should ensure a **balanced skill mix** — including design, technical, and business/strategy roles.
 - Once finalized, team members cannot be changed mid-project without prior approval from faculty.
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Deliverables

Each team is required to submit and present the following:

1. **Working Prototype / Functional Demo** of the project (web-based preferred).
 2. **Project Documentation** — including problem definition, system architecture, data flow, and AI logic.
 3. **Pitch Deck (10 slides)** summarizing:
 - The problem statement
 - Solution approach
 - Key features and innovations
 - Technology stack
 - Live demo highlights
 - Future scalability or business potential
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Presentation & Evaluation

- Each team will get **10 minutes** for the final presentation:
 - **7 minutes** for the project demo and explanation and presentation.
 - **3 minutes** for Q&A by faculty or evaluators.
 - Presentations should be **concise, visual, and startup-style**, highlighting both **technical innovation** and **real-world relevance**.
 - **All team members must participate** during the final presentation as the marking will be done on an individual basis.
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Evaluation Criteria

Evaluation Criteria		Total Marks - 20
Parameter	Description	Marks Division
Innovation & Idea Clarity	Originality and real-world applicability of the concept.	3 Marks
Technical Implementation AI/Insight Component	Functionality, AI integration, and quality of code or prototype.	7 Marks
Design & User Experience	Interface aesthetics, usability, and flow.	3 Marks
Presentation & Communication	Clarity, confidence, and team coordination during the pitch.	5 Marks
Documentation & Submission Quality	Completeness and professionalism of reports/pitch materials.	2 Marks

Capstone Project Problem Statement: Personalized E-Commerce Recommender Bot

Project Overview

1. Build an AI-powered recommendation assistant for Indian D2C food and fashion brands that delivers hyper-personalized product suggestions based on user context and preferences.
 2. The system must understand natural language queries, extract user preferences, and provide contextually relevant product recommendations with clear reasoning.
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Core Challenge: Context Engineering

To create an **AI-powered influencer marketing platform** that enables:

1. **Understands User Context:** Extracts key preferences (dietary needs, style preferences, budget) from natural language queries.
 2. **Personalized Recommendations:** Matches products to user profiles using hybrid filtering approaches.
 3. **Explains Recommendations:** Provides clear, brand-aligned reasoning for product suggestions.
 4. **Adapts to Indian Market:** Considers local pricing (₹), seasonal trends, and cultural preferences.
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Conversational Interface

Shopping Assistant: "Hi! I'm your personal shopping assistant. What are you looking for today?"

User: "I need vegan snacks under ₹300"

Shopping Assistant: "Perfect! Here are my top recommendations for you: Plant-Based Protein Cookies - ₹299

95% Match

Why it's perfect: These gluten-free cookies fit your vegan diet and budget perfectly. With 12g plant protein per serving, they're both healthy and delicious!

Quinoa Energy Bites - ₹249

87% Match

Why it's perfect: Vegan superfood snacks under your budget with natural sweetness from dates and coconut."

1. Sample Queries & Expected Outputs

User Query	Top Recommendation	Confidence	Justification
"Vegan snacks under ₹300"	Plant-Based Protein Cookies (₹299)	95%	"Perfect vegan option within budget with high protein content for health-conscious snacking"
"Light ethnic wear for summer"	Handblock Print Kurta (₹1499)	92%	"100% cotton kurta keeps you cool while maintaining traditional elegance for summer occasions"
"Protein-rich breakfast options"	Oats & Chia Seed Bars (₹349)	88%	"15g plant protein per serving makes this ideal for active lifestyles and morning nutrition"
"Trendy casual wear under ₹1000"	Oversized Cotton Tee (₹799)	85%	"On-trend oversized fit in premium cotton, perfect for casual outings within your budget"

Deliverables

- Working Demo Video (max 3 minutes):
- GitHub Repository
- PPT
- Hosted Website url(if done)